

AUSPEX: A Doctrine-Tagged, Source-Audited Corpus of State-Sponsored Cyber Operations, with a Leakage-Aware Who×Why Engine and a Pre-Registered Forecast

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Abstract

Existing cyber-conflict datasets record *who* attacked *whom*, but not the strategic *why*: the doctrine under which an operation was tasked. We present **AUSPEX**, a corpus of **785** state-sponsored (and state-adjacent criminal) cyber events, each linked to the strategic doctrines it advances, with per-claim primary sourcing, machine-verifiable schema, and a published verification methodology. Every event has been audited claim-by-claim against its raw cited sources under a documented six-point protocol; the audit reaches **100% coverage** (691 full, 94 partial). Doctrine links carry an evidentiary-confidence ladder and an *attacker/victim/defender* perspective tag, so who×why analyses use only the attacker’s own doctrine. On this corpus we report three descriptive results and one engine. First, operations are doctrinally *legible* (76% carry an attacker-rationale link) and *follow* their doctrine in time (median +3 years; 74% of dated pairs postdate publication). Second, doctrine is *informative* about attribution: null-corrected mutual information is 1.67 bits, cutting the effective suspect pool 88→28 actors. Third, deterrence is *not identifiable* from an event atlas of this size — we show why, rather than reporting a spurious effect. A leakage-aware engine attains deployed attribution top-5 of 66.1% (retrodictive) and 56.2% on a temporal holdout, and doctrine top-5 of 88.1%, decisively above a naive baseline (3.4%). Critically, the completed audit *lowered* every accuracy figure (attribution top-1 64.9→50.8): a de-circularization, not a regression. We pre-register six confirmatory predictions against post-freeze data. The corpus, engine, harness, and this paper’s figures are fully reproducible.

1 Introduction

Attribution answers *who*. For a decade, cyber-conflict scholarship and threat intelligence have converged on the who: named clusters (APT28, Lazarus, Sandworm), indictments, sanctions. But the question a defender, a policymaker, or an examiner actually needs answered is *why* — under what standing strategic frame was this operation tasked, and what does that frame predict next? Vendor attribution names the who; strategic doctrine names the why; and only the join of the two is a usable object. AUSPEX is built around that join.

*Code, data, and every validation artifact: <https://github.com/KaraZajac/AUSPEX>; frozen snapshot archived on Zenodo (DOI at release). Live atlas and engine: <https://auspex.blackflagintel.com>. Released under CC BY 4.0; layered upstream attribution in DATA-RIGHTS.md.

Concretely, AUSPEX links each cyber event to the doctrines it advances: China’s Military-Civil Fusion and 2017 National Intelligence Law; Russia’s information-confrontation and near-abroad postures; North Korea’s sanctions-evasion revenue plan; Iran’s deniable-retaliation and forward-defense doctrines; and so on across fifteen doctrine-authoring states. No prior public dataset carries these labels (§2). The contribution is four-fold:

1. **The dataset** (§3): 785 events, 222 actors, 86 doctrines (199 pillars), 1,794 per-claim source records, on a formal JSON Schema with an independent verification harness; a Datasheet for Datasets (Gebru et al., 2021) accompanies the release.
2. **A verification census** (§4): every event audited six-point against its raw sources, 100% coverage, with the status reported honestly (no “verified” overclaim) — and, unusually, we show the audit made the engine numbers *worse* on purpose.
3. **Descriptive who×why findings** (§5): doctrinal legibility, temporal precedence, doctrine→actor information, and per-state modus-operandi fingerprints — each with the confounds it does and does not survive.
4. **A leakage-aware engine and a pre-registration** (§6, §7): retrodictive and prospective attribution/doctrine prediction, and six falsifiable predictions locked before the confirmatory data exists.

We are deliberate about what the corpus is *not*: it records *disclosed* operations, not the population of operations; its ground truth is analyst-concurred vendor attribution, not a verified oracle; and its labels are analyst-mediated assessments. Every result below is framed by those limits, which we treat as first-class (§8).

2 Related work

The closest public datasets are the European Repository of Cyber Incidents (EuRepoC; EuRepoC Consortium, 2023), the Dyadic Cyber Incident and Campaign Dataset (DCID; Valeriano and Maness, 2018), the CFR Cyber Operations Tracker (Council on Foreign Relations, 2024), and the CSSM / CISSM databases. These are strong on the *who*, the *when*, and coarse effect coding; DCID in particular pioneered severity and objective coding for state dyads. What none of them carries is a *doctrine* label space — a standing strategic frame, dated to a founding document, linked to each operation with an evidentiary-confidence and perspective tag. AUSPEX’s differentiators are therefore: (i) doctrine links as a first-class label; (ii) per-claim sourcing with archived content-hashes rather than a single citation per incident; (iii) a machine-verifiable schema plus a published reliability and verification methodology; and (iv) a leakage-aware prediction engine shipped alongside the data. On the modeling side, our attribution engine follows the class-imbalance literature (Rennie et al., 2003); the honest-evaluation posture (leave-one-out, temporal holdout, null = miss, prose de-leaking) follows recent cautions on optimistic backtests in security ML.

3 The dataset

Scope and shape. AUSPEX records 785 events across fifteen doctrine-authoring states (US, RU, CN, IR, KP, IL, IN, PK, TR, BY, VN, UK, FR, KR, AE) plus a stateless **criminal** bucket

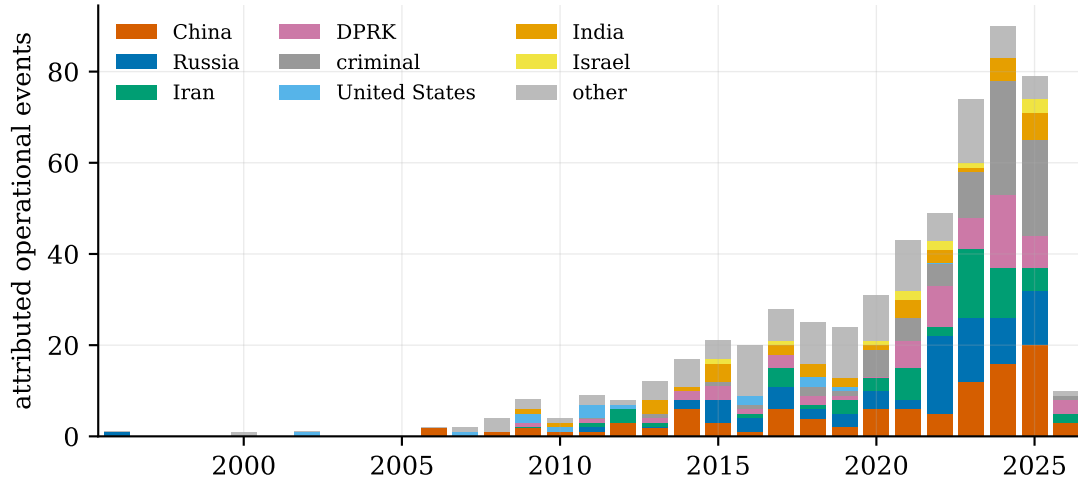


Figure 1: Attributed operational events by attacker state and year (1996–2026). Counts are collection-driven, not a census of the world; the corpus records disclosed, attributable operations. “Other” aggregates the eleven smaller doctrine-authoring states; unattributed operations are omitted from this panel.

for financially-motivated crews. Of these, 648 are *operations* (intrusion, theft, disruption, influence, pre-positioning, cyber-physical) and 137 are *meta* events (doctrine publications, sanctions and attribution notices, law-enforcement actions) — kept in the atlas and in engine training, but excluded from the prediction label sets, since predicting the doctrine of a doctrine-publication is circular. Figure 1 shows the attributed operational corpus over time: the familiar collection-driven post-2015 rise, and the shifting mix of China, Russia, Iran, DPRK, and criminal activity.

Each event. An event carries: a stable slug (never renamed; corrections add aliases); `incident_type`, `initial_vector`, dates (operation start, disclosure, end, kept distinct); attributions, each with an attributing organization, that organization’s confidence, an AUSPEX assessment, an attribution level (nation / named-actor / named-unit / activity-cluster), and a source; targets; and `doctrine_links`. Actor identifiers and aliases follow the MISP Galaxy threat-actor clustering where it exists, reviewed on conflict (MISP Project, 2024). Every load-bearing claim links to a primary source (government release, vendor report, court document, indictment); URLs are curl-verified, and where a source is un-fetchable it is recorded `url: null` with a note. We do not redistribute copyrighted source bodies — only the URL and a SHA-256 content hash (`content_sha256`) are published, so a third party can re-fetch and verify without AUSPEX re-hosting the text.

Doctrine links, confidence, and perspective. A doctrine link records the `doctrine_id`, a pillar within it, a `confidence` on a three-rung ladder, a `perspective`, and structured `inference_basis` reasoning. The confidence ladder is deliberately strict: a link is *ATTESTED* *only* when a cited source names the strategic goal in substance (with a verbatim `source_quote`); otherwise it is *strongly_inferred* or *plausible*, with the signals and ruled-out alternatives recorded. The perspective tag — `attacker-rationale`, `victim-response`, or `defender-response` — is what makes the who×why analyses honest: Stuxnet is tagged with the US/Israeli *attacker* doctrines

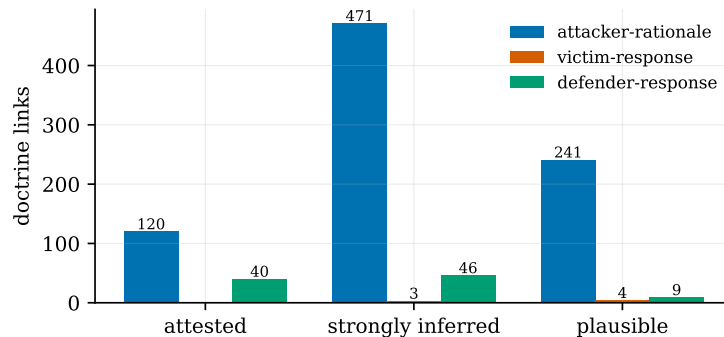


Figure 2: Doctrine links by confidence and perspective. The strict ATTESTED rung (a source names the goal) is small on purpose; who×why analyses restrict to *attacker-rationale* links and report the ATTESTED-only cut as the honest robustness check.

and, separately, with Iran’s asymmetric-warfare doctrine as a *victim-response*; only the former enter any who×why computation. Figure 2 shows the resulting distribution. The ATTESTED stratum is small by construction — 9% of operations — because it demands that a *source*, not the analyst, name the goal.

4 The verification census

The integrity backbone of the release is a human-directed, machine-executed audit of every event against its raw sources. Under a documented six-point protocol — (a) facts, (b) attribution, (c) dates, (d) doctrine links, (e) attribution level, (f) actor identity, each checked against the *raw* source text, not the atlas prose — each event was fetched, snapshotted (URL + SHA-256), corrected, and stamped. The audit is performed by a large language model (Claude Opus 4.8, recorded per event as `verified_by: claude-opus-4.8`); this is a 100%-coverage record-versus-source audit, and we are careful to disclose it as such — distinct from, and not implying, independent human inter-rater verification, which is a separate reliability study. Coverage is **100%**: 691 events *full* (every load-bearing claim on a snapshotted source) and 94 *partial* (a load-bearing source was un-snapshottable — bot-walled or link-rotted — and was mirror-corroborated); zero unstamped (Figure 3, right).

The census removed fabricated and duplicate events, merged duplicates, re-homed mis-dated events, and stripped over-attestations and fabricated tradecraft throughout. Its most instructive effect is on the engine (Figure 3, left): *every* accuracy figure fell (attribution top-1 64.9 → 50.8, doctrine 68.5 → 62.8, joint 47.9 → 38.0). This is the audit working as designed. The removed events were disproportionately the over-clean, over-attributed, or fabricated ones the engines had partly memorized; excising them *de-circularizes* the numbers. A dataset whose reported accuracy *drops* after its authors audit it is exhibiting the opposite of the optimism that inflates security-ML benchmarks.

5 Descriptive results: who×why

All results use attacker-rationale doctrine links, year granularity, and are computed by the `analysis/` scripts (`make findings`); each figure regenerates from the atlas. We report both the

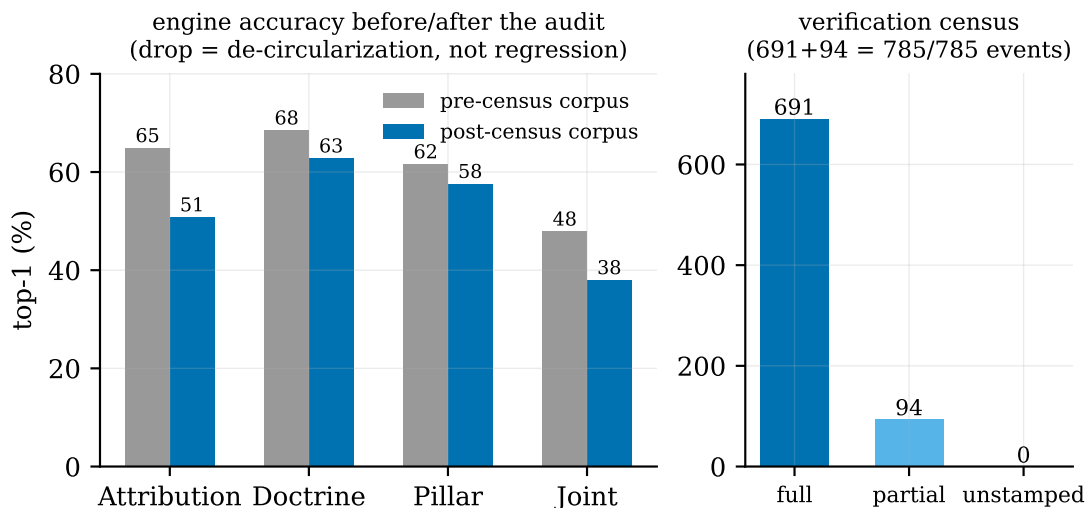


Figure 3: The verification census. *Left:* engine top-1 accuracy before and after the audit — every engine falls, a de-circularization (the audit removed over-attributed and fabricated events the model had memorized), not a regression. *Right:* 100% coverage — 691 full + 94 partial of 785, zero unstamped.

all-links and the strict ATTESTED-only cut throughout, and name the confounds each survives.

5.1 Operations are doctrinally legible and follow their doctrine

Legibility. 495 of 648 operations (76%) carry at least one attacker-rationale doctrine link. This is explicitly a *tagging-coverage* measure, not a claim about the world’s doctrinal saturation: doctrines enter the atlas partly *because* operations were observed (selection on the dependent variable), so an operation that resists a doctrinal reading may simply not get a link.

Precedence. If doctrine merely rationalized operations after the fact, operations would pre-date their documents. They do not. Across 468 (operation, dated-doctrine) pairs, the median gap is +3 years and 74% of pairs *postdate* the document’s publication (Figure 4); restricting to formally dated instruments (statute / strategy / treaty), $n = 400$ pairs, median +4 years, 72% follow. Precedence does not prove generation — a state can codify what it already does — but it rules out the simplest reverse-causation story.

5.2 Doctrine is informative about attribution

Knowing the doctrine narrows the actor. We measure this with mutual information, null-corrected by a $K = 40$ label permutation (the naive drop over-states narrowing $\sim 6\times$ on a sparse corpus, so we report only the corrected figure). Baseline actor entropy is 6.46 bits; conditioning on doctrine leaves $H(\text{actor} \mid \text{doctrine})$ such that null-corrected $MI = 1.67$ bits — 26% of actor uncertainty — cutting the effective suspect pool from 88 to 28 actors (Figure 5). The strict ATTESTED-only cut gives 0.94 bits (20%, $26 \rightarrow 14$): the same order of magnitude when a *source*, not the analyst, names the goal, so the relationship is not merely an artifact of how we chose to tag.

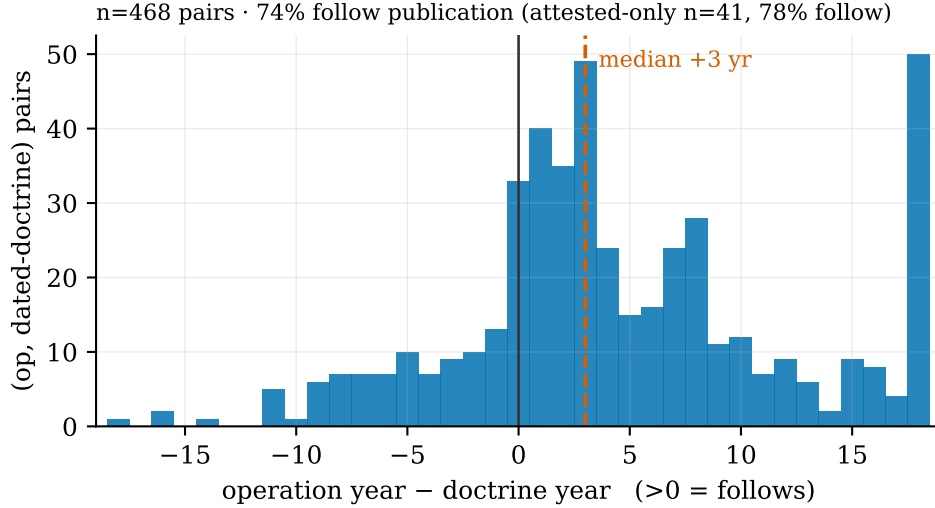


Figure 4: Temporal precedence. Distribution of (operation year – doctrine publication year) over dated pairs. The mass sits to the right of zero: operations follow their doctrine (median +3 yr, 74% follow). The small right-tail overflow is foundational doctrines (e.g. Israel’s Begin Doctrine, 1981) paired with recent operations.

5.3 States have distinct *modus operandi*

Conditioning on the terminal outcome of an operation (data theft / systems denial / money / influence / access) also carries state identity. Figure 6 shows each state’s outcome fingerprint. China is overwhelmingly data-theft (86%); the profit-motivated buckets are unmistakable — criminal 65% money, DPRK 50% money (the only *state* above 25% on money, a signature of the sanctions-evasion revenue doctrine); Russia and Iran mix data-theft with a substantial denial / disruption share (31%, 35%). Formally, $H(\text{state})$ falls from 3.54 to 3.00 bits given the outcome (null 3.44): *modus operandi* alone resolves 12% of state identity beyond chance. As an actor discriminator, outcome is real *at the tails* (DPRK money is the sharpest signal in the corpus) but weak on average, because espionage/data-theft is the universal default; a chained doctrine→target→outcome narrowing is dominated by the doctrine term (1.22 null-corrected bits) and is under-powered at 493 single-actor events.

5.4 Deterrence is not identifiable — and we say so

A natural question is whether naming-and-shaming (indictments, sanctions, attributions) deters: does an actor’s operational tempo fall after a punitive action? We ran the event study honestly — last *complete* year as the right-censoring boundary, tempo counted once per attacker-state — and the answer is that *the design cannot answer it*. The 51 usable punitive actions collapse to $n = 4$ independent (state, window) clusters; the clustered difference-in-differences is -0.69 but the sign test is non-significant ($p = 0.62$), the estimator is endogeneity-biased *toward* that sign (actions respond to surges, which then mean-revert), and the control is contaminated by other treated units (Figure 7). We therefore publish the descriptive number with an explicit refusal to read it as deterrence or defiance. The contribution here is methodological: an event atlas of this size and structure cannot identify state-level naming-and-shaming effects, and pretending otherwise is the error to avoid.

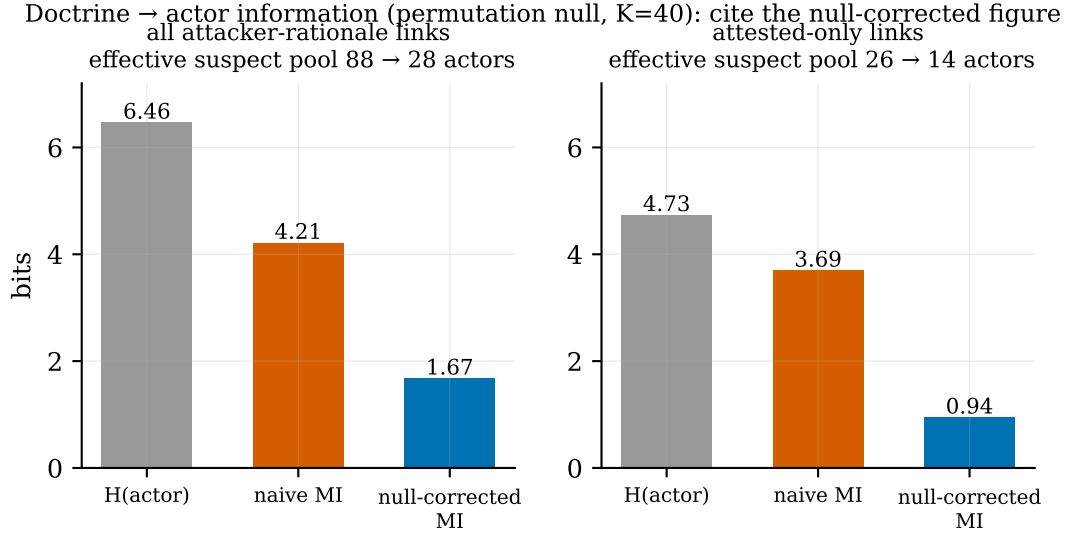


Figure 5: Doctrine→actor information. The naive mutual information (orange) is sparsity-inflated; the null-corrected figure (blue, $K = 40$ permutation) is the one to cite. Under both the all-links and the strict ATTESTED-only definition, knowing the doctrine resolves a fifth to a quarter of actor uncertainty and roughly triples the resolving power of a suspect list.

6 The engine

Four prediction engines run over the atlas: attribution (actor), doctrine, pillar, and joint (actor×doctrine). Attribution is served by a ComplementNB base (Rennie et al., 2003) — chosen for the severe class imbalance of the one-and-few-event actor tail — re-ranked by an L2 logistic stacker over discriminative meta-features (named-operator and target-org overlap, malware-lineage hits, campaign match, active-window). Doctrine and pillar use a multi-hot Naive Bayes. Features span target sectors/countries, initial vectors, incident types, MITRE ATT&CK techniques (The MITRE Corporation, 2024) and co-occurrence pairs, malware families with lineage partial-credit, named targets, geopolitical-marker proximity, dyad-reactivity lag, editorial and algorithmic campaign clusters, and TF-IDF prose — with a temporal service prior (Empirical Bayes, $\lambda = 0.2$) and temperature-scaled calibration.

Leakage discipline. Every evaluation is leakage-aware. Prose features are scrubbed of actor names/aliases *and* doctrine-identifying tokens (the analyst who wrote a summary knew the answer); a held-out event’s own algorithmic campaign membership is suppressed while it is scored; and document frequencies exclude the held-out event. We use the **null** = **miss** convention throughout: an event whose only true label is a corpus singleton, unrankable under leave-one-out, counts as a miss, not an exclusion, so numbers are comparable across engines and against the baseline.

Results. Figure 8 reports top-5 hit rates under three regimes. Retrodictive (leave-one-out / 5-fold CV) attribution reaches 66.1% top-5 (deployed ComplementNB + stack; top-1 50.8%); doctrine 88.1%, pillar 83.4%, joint 56.9%. The naive external baseline — ranking actors by a MITRE ATT&CK Groups technique lookup — scores 3.4% top-5, so the engines are learning

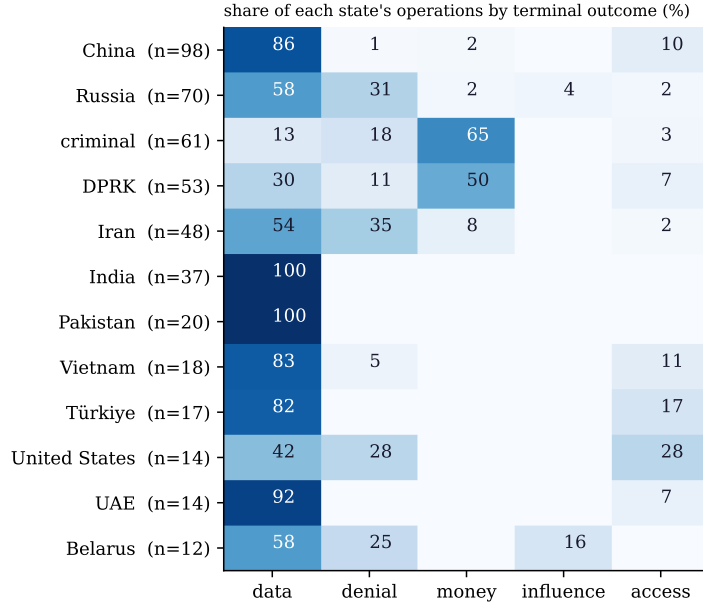


Figure 6: Per-state modus-operandi fingerprints: share of each state’s single-actor, non-false-flag operations by terminal outcome. The profit signature of **criminal** and DPRK, and the data-theft dominance of China, India, Pakistan, and the UAE, are the sharpest structure.

structure, not exploiting a trivial prior. The honest prospective figure is the **temporal holdout** (train $\leq$ 2023-12-31, score 2024+): on the *rankable* subset — events whose true actor existed before the split, the fair cold-start comparator — attribution holds 56.2% top-5 and doctrine 74.8%;¹ counting cold-start new-actor events as misses, the all-test figures are lower (attribution 40.8%), which is the genuine cost of a fast-moving corpus stated plainly. Doctrine generalizes best through time: the strategic frame an operation serves is the most temporally stable thing the engines predict, which is precisely what makes it a candidate for forecasting.

7 Pre-registration

Every result above is exploratory: discovered and tested on the same corpus. To convert the key claims into falsifiable predictions, we pre-register six confirmatory predictions about events added to the atlas *after* the frozen snapshot, evaluated under unchanged tagging rules and analysis code (Table 1). Thresholds are set deliberately below the current exploratory values to allow sampling noise while remaining falsifiable; failed predictions will be *reported*, interpreted (sampling noise vs. drift vs. claim wrong), not repaired. Two design notes reflect the completed census: P5 is a single-state claim (China’s National Intelligence Law dominance is clean; Iran’s operative doctrine is genuinely multipolar — deniable-retaliation, forward-defense, and resistance-axis run in parallel — so Iran is reported exploratorily rather than pre-registered), and P6 is scored on the rankable subset (true actor existed pre-freeze), whose current value (56.2%) clears the threshold but only narrowly, making it a genuine test rather than a lock.

¹The 56.2% is the pre-registered reference-engine (multi-hot Naive Bayes) comparator, held fixed as the P6 target; the deployed ComplementNB + stack engine scores 56.9% top-5 on the same rankable subset, so the deployed headline of $\approx$ 56% is if anything conservative.

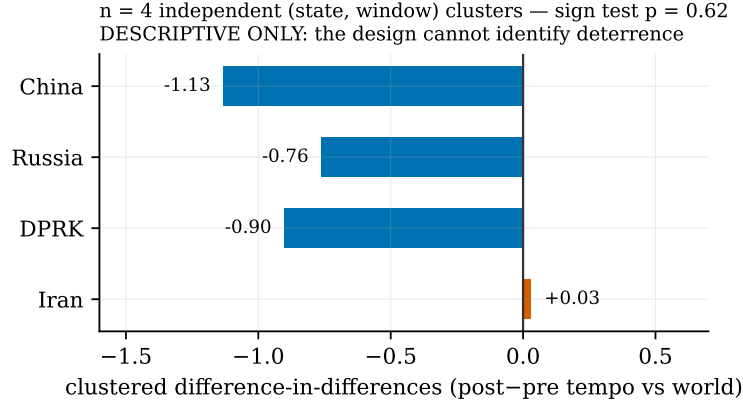


Figure 7: State-level deterrence, clustered to the effective sample. Four independent (state, window) observations; a non-significant sign test ($p = 0.62$); endogeneity biases toward apparent deceleration. Published as *descriptive only* — the design is unidentified.

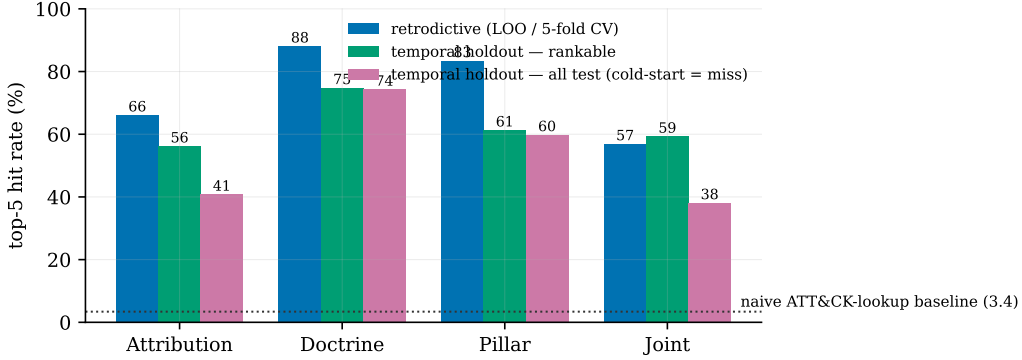


Figure 8: Engine top-5 accuracy under three regimes: retrodictive (LOO / 5-fold CV), temporal-holdout rankable (true actor seen pre-2024), and temporal-holdout all-test (cold-start counted as miss). The dotted line is the naive ATT&CK-lookup baseline (3.4%). Doctrine is the most temporally robust target; attribution pays the largest retrodictive-to-prospective gap, as expected of a corpus that adds new actors every year.

8 Limitations

Collection and disclosure bias. The corpus records *disclosed*, attributable operations, skewed toward Western reporting; absolute tempo (Figure 1) reflects collection coverage, which we measure with within-year normalization rather than assert away. **Ground-truth noise.** Attribution labels are analyst-concurred vendor assessments, not a verified oracle; every accuracy figure measures agreement with that consensus, which carries its own error rate. **Analyst-mediated labels.** Doctrine links are AUSPEX assessments; the ATTESTED-only cut is the honest one (a source names the goal), and the machine census is a record-versus-source audit, not human inter-rater verification — an inter-rater κ study is the standing next step. **Non-independence.** Events cluster by campaign and actor, so i.i.d. bootstrap intervals are somewhat narrow (estimated design effect ≈ 2); a campaign-block bootstrap is future work. **Not causal.** Legibility, precedence, and information are associations under the confounds named in §5, not evidence

#	prediction on the post-freeze cohort	threshold	current
P1	doctrinal legibility (≥ 1 attacker-rationale link)	$\geq 65\%$	76%
P2	formal precedence (op postdates dated statute/strategy/treaty)	$\geq 60\%$	72%
P3	null-corrected MI(actor; doctrine) on pooled corpus	≥ 1.0 bits	1.67
P4	DPRK financial-outcome share (only state $> 25\%$)	$\geq 30\%$	50%
P5	China’s post-pivot dominant doctrine stays #1 (nat-intel-law-2017)	persists	clean
P6	frozen engine, cold-start rankable attribution top-5	$\geq 55\%$	56.2%

Table 1: Pre-registered confirmatory predictions. “Current” is the exploratory value on the frozen 785-event corpus; thresholds sit below it. Locked at the freeze commit; the git history is the timestamp.

that doctrine *generates* operations.

9 Conclusion

AUSPEX adds the missing axis to the study of cyber conflict: the strategic *why*, as a first-class, source-audited, machine-verifiable label joined to the operational *who*. On a corpus audited to 100% coverage — an audit that made our own numbers honestly worse — operations are doctrinally legible, follow their doctrine in time, and let doctrine cut the attribution suspect pool roughly threefold; while deterrence, tested honestly, proves unidentifiable at this scale. The corpus, the leakage-aware engine, the verification harness, and every figure in this paper regenerate from public sources with a single command, and six predictions are locked against data that does not yet exist.

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